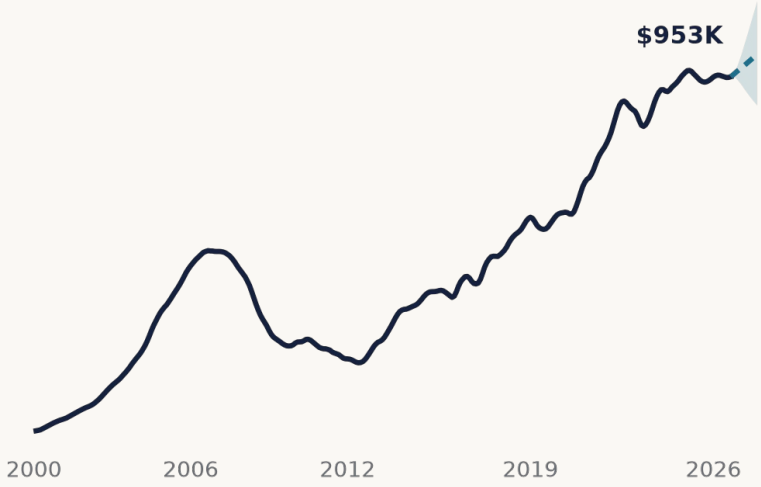
MGMTMSA 437 · FINAL PROJECT · OCTOBER 2026

Forecasting LA Home Values

How well can we forecast LA home values up to a year ahead, and which works best: time-series models or AI?

Yang Liu · Jinwoo Roh
Yiu-Hon Lee · Thanapol Doungsaeng

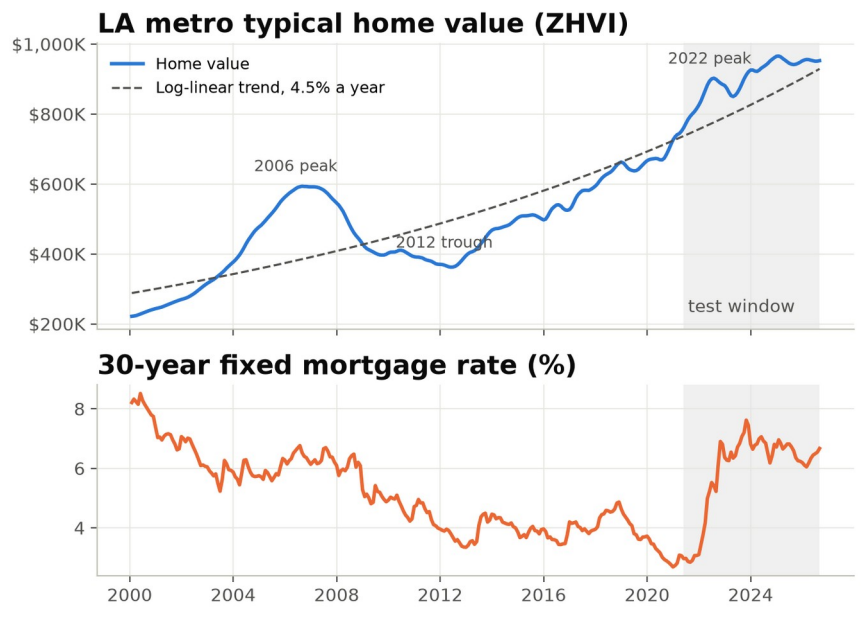
Monthly Zillow home values for the LA metro, January 2000 to August 2026, with mortgage rates and neighborhood data



Solid: actual · Dashed: our forecast to Aug 2027

WHY LA, WHY NOW

After decades of growth, LA home values have stalled



$952,601

Typical home value, LA metro, August 2026

-1.3%

Below the January 2025 peak

2.7% → 7.6%

30-year mortgage rate, 2021 low to 2023 high

DATA AND DESIGN

We tested 14 models at 18 forecast dates

DATA

Zillow ZHVI, LA metro
320 months, Jan 2000 to Aug 2026

499 LA neighborhoods
Zillow neighborhood index

Mortgage rate and payroll jobs
FRED and BLS, LA and Orange County

14 MODELS + 3 CHATBOTS

Baseline and classical
Naive drift · ARIMA · Holt · Prophet ×2

ML, deep learning, AI
XGBoost · LightGBM · LSTM · GRU · Chronos

Outside information
ARIMAX ×2 · VAR · factor VAR

Plus 3 AI chatbots
Claude · GPT · Gemini

HOW WE TESTED

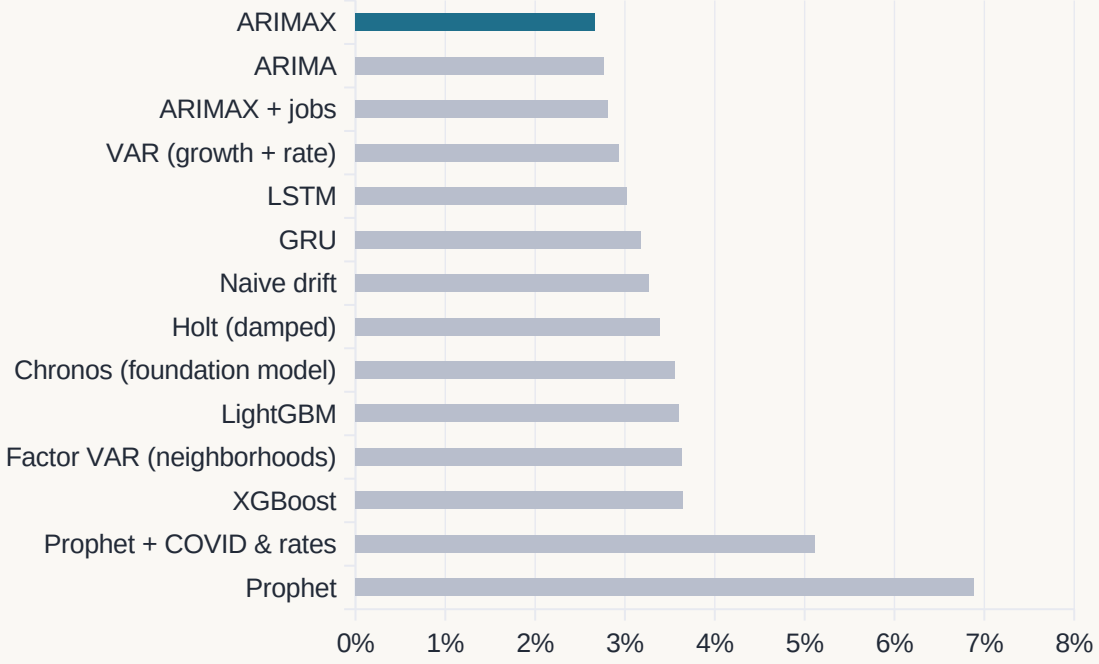
1. Rolling-origin test (main)
Repeated 12-month forecasts: update every 3 months, 18 start dates

2. Single 80/20 split
Train through April 2021, then forecast the next 64 months

HOW WE SCORE MAPE measures how far forecasts are from actual values, in percent. We average the absolute errors across months 1 to 12 and all 18 start dates. Lower is better.

RESULT 1 · ROLLING-ORIGIN TEST

ARIMAX had the lowest average forecast error



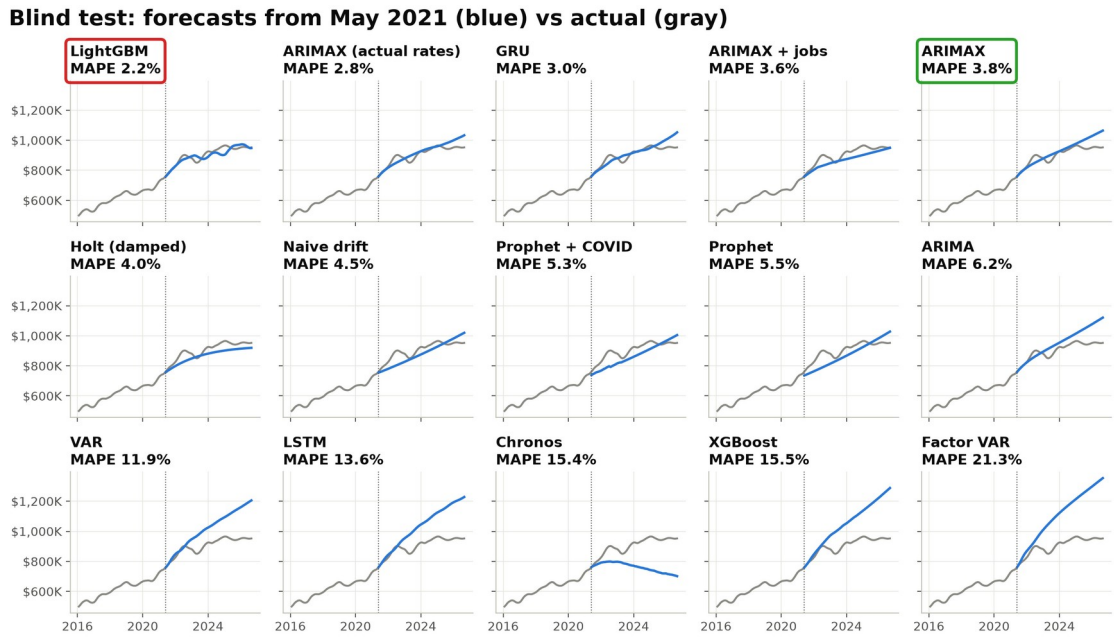
LOWEST ERROR (MAPE)

2.67%

average error across months 1 to 12. ARIMAX adds mortgage-rate information to ARIMA; plain ARIMA was close, at 2.77%. At month 12 alone: ARIMAX 5.0%, naive drift 4.9%.

RESULT 2 · SINGLE 80/20 SPLIT

LightGBM ranked first in one long test, but tenth in repeated 12-month tests



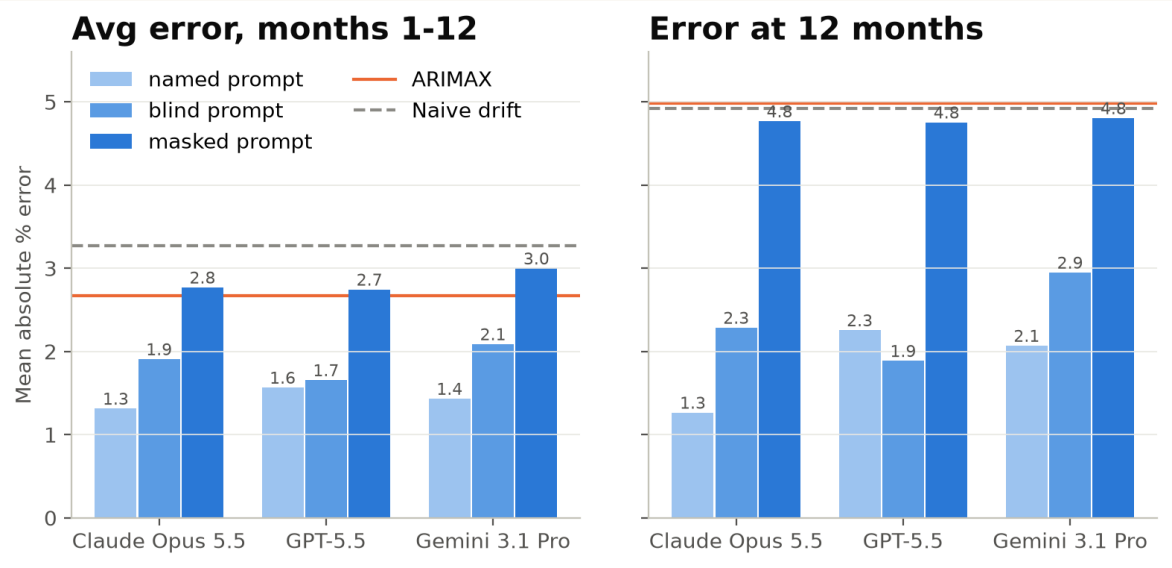
SAME MODEL, TWO TESTS

1st → 10th

1st in the 64-month test (2.2% error), 10th in repeated 12-month tests. LightGBM's lead did not hold under our main evaluation.

RESULT 3 · AI AS AN ALTERNATIVE

An alternative tool: AI's edge faded once we removed the clues



CLAUDE: NAMED → MASKED

1.3 → 2.8%

Claude's average error rose from 1.3% to 2.8%, close to ARIMAX's 2.67%. These historical periods may already be familiar to the chatbots.

RESULT 3 · AI AS AN ALTERNATIVE, HOW WE TESTED

Masking cut recognition, but didn't remove it

NAMED

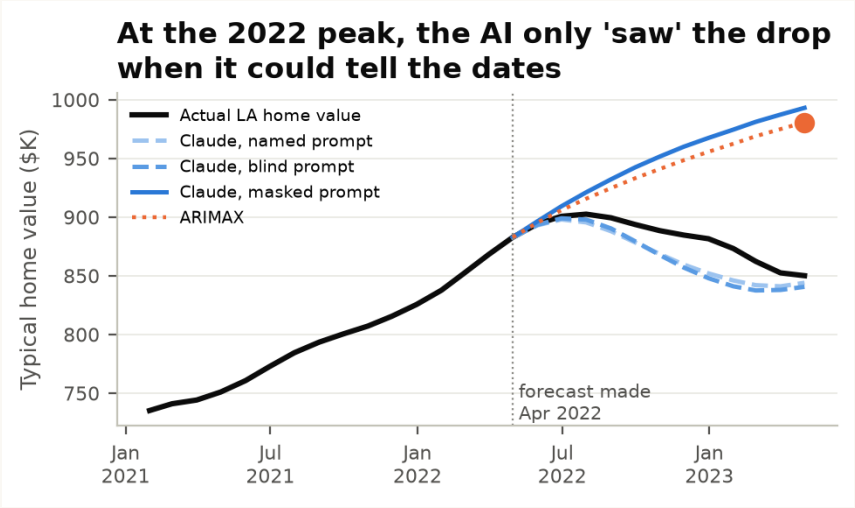
LA, real dates, dollars, rates
Most clues

BLIND

Place, dates hidden; rates kept
Rates can reveal dates

MASKED

10 years, rescaled; no place, dates, rates
Fewest clues

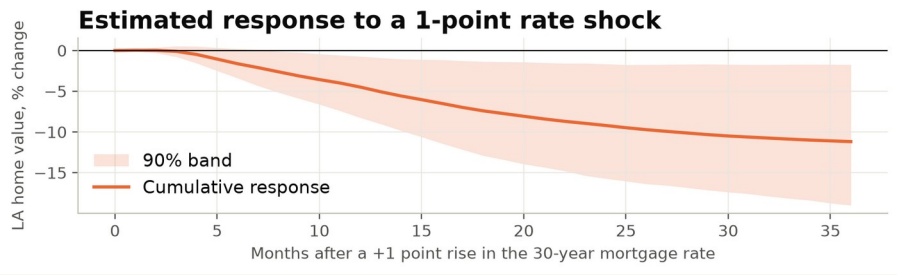


NAME THIS SERIES Claude identified LA in 16/18 blind prompts and 0/18 masked prompts. It could still estimate the period from the masked data.

RESULT 4 · MORTGAGE RATES AND OUTLOOK

Rates act with a delay; we forecast +4.8%

ESTIMATED HOME-VALUE RESPONSE TO A 1-POINT MORTGAGE-RATE SHOCK



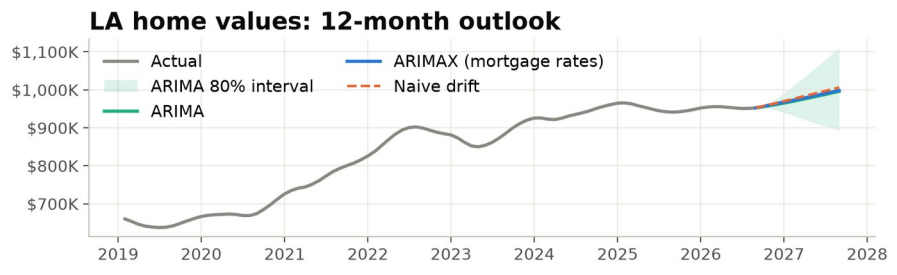
−9.2%

VAR-estimated response after two years (−4.5% after one)

p = 0.052

Granger: borderline. ARIMAX: rate p = 0.16, jobs p = 0.29

OUR FORECAST: ABOUT 5% HIGHER BY AUGUST 2027, WITH A WIDE RANGE



$998K (+4.8%)

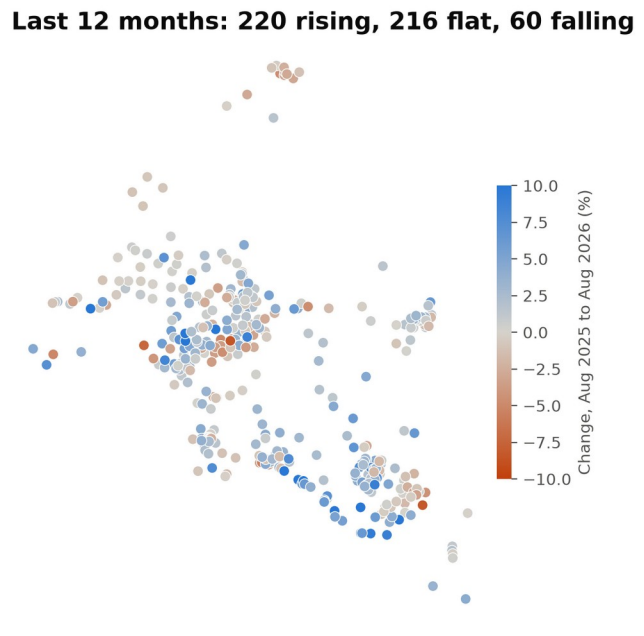
ARIMAX forecast, August 2027

-6% to +17%

for context: ARIMA's 80% interval by August 2027

NEIGHBORHOODS · LIVE MAP

Some neighborhoods are rising while others are falling



220 / 216 / 60

rising >2% · within ±2% · falling >2%, last 12 months (496 neighborhoods with recent data)

152 of 496

still below their 2022 peak, mostly in LA County

−1.2 pts

in LA County, doubling the starting home value went with a 1.2-point larger decline after the 2022 peak

Live map: neiltd.github.io/la-housing-forecast

TAKEAWAYS

Our answer: simple models held up best

01

Simple models were competitive

ARIMAX had the lowest average error; naive drift was equally competitive at month 12.

02

Test chatbots on outcomes they cannot know

Historical chatbot tests can be affected by recognition of past events.

03

Rates showed a delayed relationship

Adding jobs did not improve our forecast; turning points stayed hard to call.

04

The metro average is only part of the picture

Results varied; per-neighborhood ARIMA beat naive drift (2.8% vs 3.5%).

NEXT STEPS Test more cities; score the AI live on months after its training cutoff; feed AI altered histories to check whether it forecasts the data or recalls the past.